# HOVER: Part II of the Scientific Proposal

## 1. Preliminary results and further background

*(Fig. B1: MESH contact map and EM cut-outs; electrical wiring diagram and RMN tuning prediction; Tier-1 traces; micro-CT of the male MIO; one controversial stimulus.)*

**P1 MESH: electrical synapses at connectome scale.** MESH retrieves the surface meshes of selected FlyWire neurons, builds a k-d tree over mesh vertices and finds every location where two membranes lie within 100 nm. Contacts are grouped into sites by hierarchical clustering, validated in the 8-nm EM volume (only genuine membrane appositions without an annotated chemical specialisation are kept) and served to experts in a viewer. For the 26-neuron set of HS, VS and retinal motor neurons (RMNs) in both hemispheres it found 619 sites, of which 49 motor-neuron–LPTC sites survived validation (2,895 EM images, public viewer), including the best-characterised candidate, between the MOT motor neuron and HSN [14,41].

**P2 The electrical gaze pathway.** Each retinal muscle is innervated by one motor neuron (MOS RMN: FlyWire type CB0901; MOT RMN: CB0804). The two muscles pull orthogonally (51° and 320°). Only 3 % of the RMNs' chemical input comes from visual neurons, but neurobiotin fills and MESH agree that the MOS RMN is coupled to the three HS cells and VS1–4, and the MOT RMN to the HS cells only. A linear sum over the coupled cells predicts each RMN's preferred direction within 8°, and the vector sum of muscle pulls weighted by RMN activity predicts the two-dimensional optokinetic reflex ($R^2 = 0.994$). Mutants lacking the neuronal gap-junction protein shakB show a reduced reflex, and spontaneous retinal movements are preceded by motor-related potentials in HS cells that mirror the visual consequence of the movement ($R^2 = 0.97$, slope $-1$) [14]. The pathway from photoreceptor to retinal movement is therefore complete at the level of identified cells, which is the precondition for O1.

**P3 Tier-1 model.** A Hodgkin–Huxley model of the right-eye circuit, with electrical and chemical coupling, runs inside MESH. It reproduces sustained RMN spiking and the eye-displacement direction observed after optogenetic activation of the MOT and MOS motor neurons (unpublished).

**P4 Differentiable detectors and controversial stimuli (ALIstim).** A PyTorch bank of Hassenstein–Reichardt and Barlow–Levick detectors with ON/OFF pathways, motion opponency [26] and wide-field pooling, with which stimuli are optimised by gradient descent for one model or for the difference between two. An open NumPy reimplementation adds gap-junction-coupled LPTC → RMN circuits for *Drosophila* and for male and female *Eristalis* (including HSNE and the male MIO) and already returns stimuli that separate competing male-circuit hypotheses (proof of principle, placeholder parameters; `code/` in the project repository).

**P5 Comparative anatomy.** Micro-CT and synchrotron tomography of 167 insect species, behaviour in 59, show specific retinal muscles only in Diptera. Hoverflies are among the earliest-branching flies with a specific retinal muscle (MOT) and an optokinetic reflex, and males carry the additional M. interocularis (MIO) between the two eyes, placed to shift the bright-zone retina [2]. A thinner circumocular ridge is associated with retinal movements across the phylogeny, and both known muscles probably derive from antennal musculature: the first clues to where the MIO comes from.

**P6 Edge AI, teaching and participation.** TinyML classifiers for real-time event detection on Arduino-class hardware (Computational SpikerBox, SpikerBot; NIH SBIR), a high-school AI and neuroscience curriculum (book and code, in progress), a participatory plant-electrophysiology study with pupils as first authors [88], a research fellowship co-directed in more than 20 countries, and IU computer-vision courses from [semester; fill] with [n] students per cohort.

## 2. Methodology and work plan

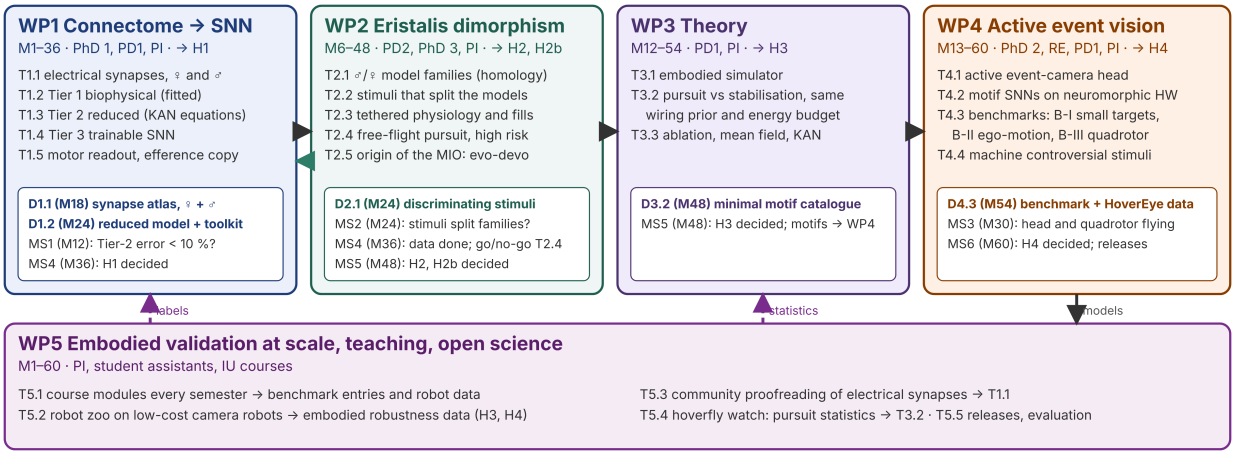



**Figure B2.** Work packages, tasks, deliverables (D), milestones (MS) and the main dependencies. Solid arrows: models and motifs flow from WP1 through WP3 to WP4 and onto robots; dashed arrows: data, labels and task statistics flow back from WP2 and WP5.

Five work packages share one modelling ladder and one stimulus engine (Fig. B2). "→ H" marks the hypothesis a task tests; D = deliverable; M = project month. Team: PI; Postdoc 1 (PD1, IU, computational neuroscience); PhD 1 (IU, connectomics and machine learning); PhD 2 (IU, event-based spiking vision); research engineer (RE, IU, neuromorphic robotics, part-time); Postdoc 2 (PD2, MPI for Biological Intelligence, hoverfly physiology); PhD 3 (LMU, evolutionary developmental genetics); student assistants (SA).

### WP1 Connectome → spiking circuit, with electrical synapses (M1–36; PhD 1, PD1, PI) → H1

- **T1.1 Electrical synapses of the gaze system in both sexes.** Neuron set: all lobula-plate tangential cells (the HS, VS and the further types catalogued in [90]), the four RMNs, neck motor neurons and optic-flow descending neurons, in the female FlyWire connectome (v783 [12]) and the male central-nervous-system connectome (v1.0 [39]). MESH detects appositions; a 3D convolutional classifier trained on the expert-validated sites (membrane apposition, pentalaminar profile, no chemical specialisation; negatives are rejected sites) ranks the rest; uncertainty-ranked queues go to expert and community proofreading (T5.3). *Validation:* held-out expert labels (precision and recall), agreement with the neurobiotin coupling in [14] and with the innexin immunostaining maps of [50]. *Sex comparison:* cell types are matched across the two connectomes with the cross-dataset typing of [47,39]; chemical and electrical connectivity of the gaze circuit are compared type by type. The expected result, a monomorphic circuit in *Drosophila*, is the baseline against which the hoverfly dimorphism is read; a difference would itself be a finding for H1's corollary. **D1.1 (M18):** open atlas of candidate electrical synapses of the fly gaze system, female and male, with the first sex comparison of this circuit (corollary of H1).
- **T1.2 Tier 1, biophysical.** Morphologically reduced multi-compartment HS, VS and RMN models built from the EM skeletons, active spike generation in the RMNs, chemical synapses from the connectome and gap-junction conductances proportional to validated apposition area. Parameters are fitted by gradient descent in a differentiable simulator [66] to published HS/VS physiology [11,31,32], RMN calcium tuning and the optogenetic displacement vectors of [14]. The front end is the differentiable detector bank (P4), with the connectome-constrained optic-lobe model of [45] as an alternative input.
- **T1.3 Tier 2, reduced.** Integrate-and-fire or adaptive-exponential point neurons with linear gap-junction coupling. Effective couplings come from Tier-1 transfer impedances. For each module (dendritic integration, gap-junction transfer, muscle mechanics) small Kolmogorov–Arnold networks [93] are trained on Tier-1 input–output pairs, pruned and regressed to closed-form expressions, so that the reduction is explicit and inspectable. We report error bounds on RMN tuning and latency (target < 10 % against Tier 1). **D1.2 (M24):** reduced model and the reduction toolkit, reusable for other circuits.
- **T1.4 Tier 3, trainable.** A connectome-masked SNN with sign constraints, trained with surrogate gradients [64,65] in a JAX/PyTorch stack that exports to the neuromorphic toolchains used in WP4. Gap junctions enter as differentiable linear couplings, which also regularises training.
- **T1.5 Motor readout and efference copy.** Muscle activation → force (Hill-type) → retinal displacement through the measured pull directions; the efference copy is a model output, tested against the HS recordings in [14]. Validated against retinal tracking in WP2.

**WP2 The dimorphic gaze circuit of *Eristalis* (M6–48; PD2 at MPI-BI, PhD 3 at LMU, PI) → H2, H2b**

- **T2.1 Model families.** Male and female circuits are built by homology (HSN, HSE, HSS, HSNE, VS; RMNs of MOT, MOS and MIO), constrained by the tomography atlas (P5), the fills of T2.3 and published hoverfly physiology [1,15,17,23]. Four male hypotheses: **M-H2** (HSN → MIO), **M-HSNE** (HSNE → MIO), **M-HS** (pooled HS → MIO), **M-VS** (VS → MIO), each with and without an additional small-target input [16,63].
- **T2.2 Controversial stimuli.** For every model pair, stimuli are optimised to maximise the predicted difference in retinal trajectory or MIO activity, within display limits (luminance, contrast, spatio-temporal frequency) and under a naturalness prior. The expected evidence per trial sets the number of animals: the **power analysis is done in silico before any experiment**. **D2.1 (M24):** the stimulus set.
- **T2.3 Experiments (MPI for Biological Intelligence, Fenk laboratory; PD2, M7–42).** Tethered male and female *Eristalis*: retinal tracking through the cornea by deep-pseudopupil imaging under infrared light [7,14]; electromyography of MOT, MOS and MIO (the MIO is identified by its position between the two eyes [2]); intracellular recordings and neurobiotin fills of HSN, HSNE and the RMNs with sharp electrodes, the standard method in hoverfly tangential cells [1]; visual stimulation on an open platform [22]; animals reared year-round [19]. *Analysis:* hierarchical Bayesian comparison of the model families across animals [35].
- **T2.4 Free-flight retinal tracking (high risk, high gain).** Pursuit of artificial targets in an indoor arena [21] with head-mounted ultralight markers and high-speed cameras. Started only after T2.3 succeeds (decision at MS4); no other task depends on it.
- **T2.5 Developmental and genetic origin of the MIO (LMU, R. Merrill; PhD 3, M7–48) → H2b.**
  - *Origin:* a developmental series of male and female pupae (micro-CT and phalloidin confocal imaging) identifies the anlage the MIO forms from. Candidates: a split of the MOT anlage, or a further repurposed antennal muscle, as proposed for MOS and MOT [2].
  - *Genes:* sex- and stage-resolved RNA-seq of the anlage region at three pupal stages (pooled, low-input protocols), using the chromosome-level *E. tenax* genome [44]; male-biased muscle-identity genes and the sex-determination factors *doublesex* and *fruitless*; candidates localised by in situ hybridisation chain reaction in the muscle and in the MIO motor neuron.
  - *H2b, the motor neuron specifies the muscle:* in *Drosophila* the male-specific Muscle of Lawrence forms only when its motor neurons carry the male *fruitless* programme [42,43,79]. We predict the same logic in *Eristalis*: a male-specific identity of the MIO motor neuron induces the muscle, so that one dedicated neuron builds one dedicated muscle, mirroring H2.
  - *Comparison:* syrphid species with and without a male MIO from the tomography survey test whether candidate changes track the trait. *Exploratory, high risk:* RNAi knock-down in pupae.

**WP3 Theory: why is vision dimorphic? (M12–54; PD1, PI) → H3**

- **T3.1 Embodied simulator.** Compound-eye sampling (including the male bright-zone resolution [15]), retinal actuators, simplified flight dynamics; GPU-based and open.
- **T3.2 Task optimisation.** Tier-3 networks are trained on *pursuit* (small fast targets in clutter [16,21]) or on *stabilisation* (hovering against optic flow; feeding approach [18]). Both conditions share the wiring prior and the spike-count energy penalty [80], and both may recruit a third actuator and shape its movement policy. Readouts: receptive-field width and position of HSN-like units, actuator recruitment and the performance–energy trade-off. Controls: shuffled wiring prior, a range of energy penalties, ≥ 10 seeds, and a variant in which the sensor-movement policy is optimised jointly with the network.
- **T3.3 Analysis.** Ablation to the minimal motif set; linear-response and mean-field theory of gap-junction-coupled populations; information-theoretic receptive-field analysis; Kolmogorov–Arnold readouts that convert trained networks into equations. **D3.2 (M48):** the motif catalogue.

**WP4 Active event-based machine vision (M13–60; PhD 2, RE, PD1, PI) → H4**

- **T4.1 Active event-camera head.** A high-resolution event camera on a two-axis piezo tip-tilt stage that reproduces fly retinal trajectories (amplitudes of a few degrees [7]), carried by a pan-tilt gimbal on a motion platform that replays ego-motion. Movement policies: none, fixed artificial microsaccades [54], fly-derived (recorded in WP2, including the male MIO pattern) and learned (T3.2). Measures: event yield, static-scene acuity and latency.

- **T4.2 Motif SNNs on neuromorphic hardware.** Catalogue motifs (D3.2) become event-driven SNNs, deployed on a sensor-integrated chip with sub-milliwatt resting power [70] and on large-scale systems [67,68] through EBRAINS and vendor research programmes. Ablations: a Kolmogorov–Arnold parameterised event front end against convolutional front ends.
- **T4.3 Benchmarks (civil tasks, pre-registered protocols).**
  - **B-I Small-target detection and tracking in clutter:** our own *HoverEye* dataset, recorded with the active head (balls, insects and small drones against vegetation, with motion-capture ground truth) plus the public EV-IMO set [60]. Baselines: insect-inspired target detectors [61,107], event-based deep detectors [94] and converted SNNs. Metrics: detection F1, track continuity, event-to-output latency.
  - **B-II Ego-motion and optic-flow estimation:** MVSEC [59] and DSEC [58]. Baselines: E-RAFT [57], spiking optic-flow networks [55,56] and the fly-derived velocity CNN [38]. Metrics: endpoint error, latency, synaptic operations and measured energy on hardware against a GPU board.
  - **B-III Closed-loop hovering and pursuit on a quadrotor:** an indoor event-camera quadrotor with an on-board neuromorphic processor in a netted motion-capture arena, the robotic counterpart of the hoverfly arena of [21]. Tasks: hold position against optic flow (female task) and intercept a moving target (male task). Baselines: classical flow-based control and a GPU CNN pipeline [55]. Metrics: position error, time to capture, energy per trial.

  *Protocol*. Each benchmark is pre-registered (tasks, splits, metrics, baselines, seeds, stopping rules) before data collection. Models are matched in parameter count and in synaptic operations per inference; energy is measured with a power meter on the deployed hardware and reported as accuracy per joule next to the GPU reference; five seeds per model; HoverEye is recorded under a CC-BY licence with target sizes, speeds and backgrounds drawn from the hoverfly-watch statistics (T5.4). H4 counts as supported if the motif SNNs with active sampling reach at least the baseline accuracy at one third of the energy, or 20 % higher accuracy per joule, on two of the three benchmarks. Results go to an open leaderboard that accepts outside entries. **D4.3 (M54):** the benchmark, HoverEye and the `hover-snn` library.
- **T4.4 Controversial stimuli for machines.** The WP2 engine applied to CNN–SNN pairs as a robustness diagnostic, and to the three movement policies.

**WP5 Embodied validation at scale, teaching and open science (M1–60; PI, SA, IU courses)**

- **T5.1 Course integration.** HOVER modules in IU computer-vision courses every semester (see the teaching plan in the repository); student projects deliver benchmark entries and robot data; distance students receive a robot kit by mail.
- **T5.2 Robot zoo.** Schools, makerspaces and citizens run fly-type, female-type and male-type circuits on low-cost two-wheeled camera robots over printable pursuit and stabilisation courses. Logs (spikes, motor commands, camera-derived performance) go to an open database and are analysed with mixed-effects models [35] with participant and device as random effects: thousands of embodied trials under uncontrolled optics, lighting and noise.
- **T5.3 Community proofreading** of candidate electrical synapses in the MESH viewer, with an expert gold standard, feeding T1.1 [85].
- **T5.4 Hoverfly watch.** Smartphone slow-motion videos of male hovering and pursuit in gardens, with location and EXIF data stripped, give natural pursuit statistics (target sizes, speeds, backgrounds) for T3.2 and B-I.
- **T5.5 Open releases and evaluation.** Code under MIT, data under CC-BY, teacher kits; firmware hardware-neutral with one commercial and one open reference platform; learning outcomes measured before and after; IU ethics approval.

*Conflict of interest:* the PI holds a senior role at Backyard Brains, the manufacturer of one of the candidate classroom robots. IU procurement selects robots against a hardware-neutral specification without the PI's involvement, the quadrotor and the active head (WP4) use unrelated vendors, and all code runs on at least one open platform.

## 3. Timeline, milestones and decision points

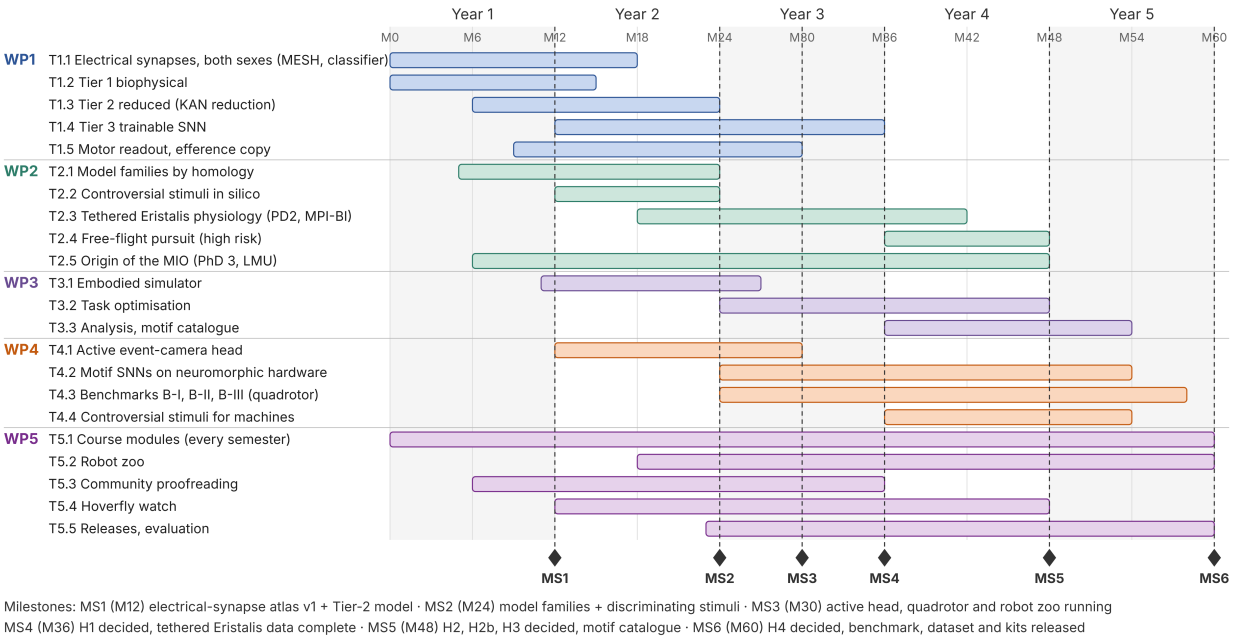



**Figure B3.** Gantt chart of the 22 tasks and six milestones (M1 = September 2028, placeholder). WP1, WP3 and WP4 form the modelling and engineering spine; WP2 and WP5 run alongside and feed data back.

| MS | Month | Milestone | Decision |
|---|---|---|---|
| MS1 | 12 | Electrical-synapse atlas v1 (female); Tier-2 *Drosophila* model | Tier-2 error against Tier 1 < 10 %? If not, Tier 1 serves WP2 |
| MS2 | 24 | Male atlas and sex comparison; male and female *Eristalis* model families; controversial stimuli designed in silico | Do the stimuli separate the families with the planned number of animals? If not, refine models before experiments |
| MS3 | 30 | Active head and quadrotor flying; robot zoo launched; HoverEye v1 | Hardware access confirmed, else GPU emulation with energy modelling |
| MS4 | 36 | H1 decided; tethered *Eristalis* data complete | Constraints fixed for WP3; go/no-go for T2.4 |
| MS5 | 48 | H2, H2b and H3 decided; motif catalogue | Motifs passed to WP4 |
| MS6 | 60 | H4 decided; benchmark, models, dataset and kits released | — |

## 4. Hypotheses: tests and what each outcome teaches

| Hyp. | Supported if | Falsified if | Value if falsified |
|---|---|---|---|
| H1 | The Tier-2 SNN with EM-derived weights predicts RMN preferred directions (± 15°), eye-movement direction and latency in both sexes | Connectivity-only models miss ≥ 2 RMN tunings | Unannotated or neuromodulatory inputs matter; the MESH search is widened |
| H2 | MIO activity follows HSN-preferred motion confined to the dorso-frontal field | MIO EMG tracks MOT-type or VS-type motion | The muscle is a general range extension; the model comparison still decides between families |
| H2b | *fruitless* or *doublesex* are expressed male-specifically in the MIO motor neuron and anlage; the muscle forms from a defined anlage only in males | No sex-biased expression in the motor neuron; specification is muscle-intrinsic | A muscle-intrinsic programme; still the first developmental account of a dimorphic eye muscle |
| H3 | Narrowed dorso-frontal integration and an extra effector emerge only under the pursuit objective | They emerge under both objectives, or neither | Points to developmental or genetic constraint rather than task optimality |
| H4 | Motif SNNs with active micro-movements beat matched baselines in accuracy per energy on ≥ 2 of 3 benchmarks | No advantage | A rigorous, open negative result for bio-inspired claims; the benchmark remains |

## 5. Risks and mitigation

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Gap-junction classifier does not generalise beyond validated sites | Medium | Medium | Expert and community proofreading; gap-junction weights become bounded free parameters |
| No *Eristalis* connectome, so the homology mapping is wrong | Medium | High | Competing model families instead of one model; discriminating stimuli; fills as ground truth |
| Hoverfly physiology slower than planned (non-genetic animal) | Medium | High | An experienced postdoc at the only site with the set-ups; stimuli designed in silico first; WP1, WP3 and WP4 do not wait |
| Developing muscle tissue too small for RNA-seq | Medium | Medium | Pooled samples, low-input protocols; HCR for *Drosophila*-known candidates as fallback |
| Free-flight tracking fails | High | Medium | Tethered closed-loop virtual flight; no task depends on T2.4 |
| Quadrotor integration delayed or unsafe | Medium | Medium | Benchmarks B-I and B-II are independent; the active head on the motion platform covers ego-motion; indoor netted arena, no outdoor flight |
| Neuromorphic hardware access restricted | Low | Medium | Two hardware routes plus GPU emulation with calibrated energy models |
| H3 null | Medium | Low | Interpretable result (constraint versus optimality) |
| SNNs lose to deep baselines (H4 null) | Medium | Medium | Reported rigorously; hybrid motif front end with learned back end as a secondary result |
| Recruitment delays | Low | Medium | Posting on EURAXESS at signature; IU computer-vision students as the MSc → PhD pipeline |

## 6. Team, working arrangements and environment

- **PI (50 %, above the 40 % minimum):** scientific lead; WP1 and WP3 modelling; WP5 lead.
- **Team, three beneficiaries.** At IU: PD1, computational neuroscience and SNN theory (48 months); PhD 1, connectomics and machine learning; PhD 2, event-based spiking vision and neuromorphic hardware; a part-time research engineer for the active head, the quadrotor and the robot fleet (M13–48); student assistants. At MPI for Biological Intelligence (Active Sensing, L. Fenk): PD2, hoverfly physiology (36 months, M7–42). At LMU Munich (R. Merrill): PhD 3, evolutionary developmental genetics (48 months). The PI selects, funds and supervises all team members; the partner group leaders co-supervise locally and give access to their set-ups.
- **Relation to the Fenk laboratory's ERC project** on *Drosophila* eye-movement mechanisms: HOVER works on *Eristalis*, on models and on machines; the division of scope is fixed in a written collaboration agreement; data and tools flow both ways.
- **Doctoral route:** PhD students enrol at LMU (doctorate-granting), co-supervised by the PI [confirm the agreement and whether IU has doctoral rights].
- **Collaborators** (written consent obtained before submission; roles limited to what is stated): L. Fenk (physiology set-ups, *Drosophila* RMN data, tomography access); R. Merrill (genomics and developmental genetics); [neuromorphic hardware platform contact]; [drone or event-camera partner].
- **Working arrangements:** monthly all-team meeting in Munich; quarterly WP reviews against the milestones; one shared code and data repository with continuous integration; pre-registration of the benchmark protocols; a data management plan by M6.
- **Data and software management:** every model tier, stimulus set, recording and benchmark run is versioned and released with a DOI (Zenodo, DANDI for physiology, the FlyWire and MaleCNS annotation channels for proofread synapses); software follows a tested, documented release process with the pre-built MESH viewer as the model; PhD 1 acts as data steward with 10 % of time; a FAIR audit is part of every milestone review.
- **Environment:** the IU computer-vision group, its students and GPU resources [confirm]; physiology set-ups and tomography at MPI-BI; genomics infrastructure at LMU; EBRAINS neuromorphic access.

## 7. Ethics, civil focus, open science and data

- **Civil focus:** HOVER has an exclusive focus on civil applications. The quadrotor flies indoors only. Insects only (no vertebrates). Human participants (students, citizens including school pupils) take part with informed and parental consent under IU ethics approval, with GDPR-compliant data handling; uploaded videos are stripped of location and metadata.
- **AI:** HOVER develops AI-based vision models; no personal data are used for training.

- **Open science:** code under MIT, data under CC-BY via DANDI and Zenodo, immediate open access to publications, pre-registered benchmark protocols. Details are in the ethics self-assessment.

---

## Funding ID (not counted in the page limit)

*Required: current research grants and their subject, and any ongoing application for work related to this proposal (IfA p. 29).*

| Title | Funder | Amount | Period | Role | Status | Relation to HOVER |
|---|---|---|---|---|---|---|
| [IU start-up package] | IU | [€] | [ ] | PI | [current] | [what it pays for; no overlap with HOVER costs] |
| [LMUexcellence Innovation Fund project] | LMU | [€] | [2024–] | [role] | [status] | [subject; no overlap] |
| [other current grants] | | | | | | |

No applications for work related to HOVER are pending.

## References

See `references.md` (not counted in the page limit).

## References

Numbering 1–44 follows the DFG proposal so that figure and text citations carry over; 45 onwards were added for the ERC version. Applicant publications are in **bold**. Entries marked [verify] could not be checked against the publisher's page from this environment; check them before submission. Author lists must appear exactly as published (IfA p. 28).

[1] Nordström K, Barnett PD, Moyer de Miguel IM, Brinkworth RSA, O'Carroll DC (2008). Sexual dimorphism in the hoverfly motion vision pathway. Curr Biol 18(9):661–667. **[2] Kviatko P, Renous Y, Falt T, Kroell LM, Guerreiro Mota S, Pujari A, Rimniceanu M, Serbe-Kamp E, Šimsová E, Vollrath F, Wilde F, Hammel JU, Scheel M, Blanke A, Fenk LM (2026). Spontaneous eye movements preceded visual stability reflexes in the evolution of insects. Manuscript [status: submitted / in preparation; post to bioRxiv before 12 Jan 2027 and add the DOI; verify].** [3] Franceschini N, Chagneux R (1997). Repetitive scanning in the fly compound eye. In: Proceedings of the 25th Göttingen Neurobiology Conference (Wässle H, Elsner N, eds). Thieme. [4] Franceschini N (1994). Combined optical, neuroanatomical, electrophysiological and behavioural studies on signal processing in the fly compound eye. In: Biocybernetics of Vision: Integrative Mechanisms and Cognitive Processes (Taddei-Ferretti C, ed). World Scientific, pp. 341–355. [5] Franceschini N (2014). Small brains, smart machines: from fly vision to robot vision and back again. Proc IEEE 102(5):751–781. [6] Viollet S (2014). Vibrating makes for better seeing: from the fly's micro-eye movements to hyperacute visual sensors. Front Bioeng Biotechnol 2:9. **[7] Fenk LM, Avritzer SC, Weisman JL, Nair A, Randt LD, Mohren TL, Siwanowicz I, Maimon G (2022). Muscles that move the retina augment compound eye vision in *Drosophila*. Nature 612(7938):116–122.** [8] Kim AJ, Fenk LM, Lyu C, Maimon G (2017). Quantitative predictions orchestrate visual signaling in *Drosophila*. Cell 168(1–2):280–294.e12. [9] Burtt ET, Patterson JA (1970). Internal muscle in the eye of an insect. Nature 225:563–564. [10] Hengstenberg R (1971). Das Augenmuskelsystem der Stubenfliege *Musca domestica*. I. Analyse der "clock-spikes" und ihrer Quellen. Kybernetik 9:56–77. [11] Haag J, Borst A (2004). Neural mechanism underlying complex receptive field properties of motion-sensitive interneurons. Nat Neurosci 7(6):628–634. [12] Dorkenwald S, Matsliah A, Sterling AR, Schlegel P, Yu S-C, McKellar CE, Lin A, Costa M, Eichler K, Yin Y, et al. (2024). Neuronal wiring diagram of an adult brain. Nature 634:124–138. [13] Zheng Z, Lauritzen JS, Perlman E, Robinson CG, Nichols M, Milkie D, et al. (2018). A complete electron microscopy volume of the brain of adult *Drosophila melanogaster*. Cell 174(3):730–743.e22. **[14] Falt T*, Ammer G*, Serbe-Kamp É, Kroell LM, Friedrich AB, Guerreiro-Mota S, Šimsová E, Fenk LM (2026). Vectorial efference copy and visuomotor transformation through gap junctions. bioRxiv, doi:10.64898/2026.08.11.744090 (posted 17 Aug 2026). *Equal contribution.** [15] Straw AD, Warrant EJ, O'Carroll DC (2006). A 'bright zone' in male hoverfly (*Eristalis tenax*) eyes and associated faster motion detection and increased contrast sensitivity. J Exp Biol 209(21):4339–4354. [16] Nordström K, Barnett PD, O'Carroll DC (2006). Insect detection of small targets moving in visual clutter. PLoS Biol 4(3):e54. [17] Nicholas S, Leibbrandt R, Nordström K (2020). Visual motion sensitivity in descending neurons in the hoverfly. J Comp Physiol A 206(2):149–163. [18] Thyselius M, Gonzalez-Bellido PT, Wardill TJ, Nordström K (2018). Visual approach computation in feeding hoverflies. J Exp Biol 221(10):jeb177162. [19] Nicholas S, Thyselius M, Holden M, Nordström K (2018). Rearing and long-term maintenance of *Eristalis tenax* hoverflies for research studies. J Vis Exp (135):e57711 [verify article number]. [20] Ogawa Y, Nicholas S, Thyselius M, Leibbrandt R, Nowotny T, Knight JC, Nordström K (2023). Descending neurons of the hoverfly respond to pursuits of artificial targets. Curr Biol 33(20):4392–4404.e5. [21] Thyselius M, Ogawa Y, Leibbrandt R, Wardill TJ, Gonzalez-Bellido PT, Nordström K (2023). Hoverfly (*Eristalis tenax*) pursuit of artificial targets. J Exp Biol 226(4):jeb244895. [22] Prech S, Groschner LN, Borst A (2024). An open platform for visual stimulation of insects. PLoS ONE 19(4):e0301999. [23] Nicholas S, Sporar Klinge K, Turnbull L, Moran A, Young A, Ogawa Y, Nordström K (2026). Sexual dimorphism in sensorimotor transformation of optic flow. eLife 15:RP109795 (reviewed preprint; bioRxiv doi:10.1101/2025.10.16.682952) [verify version]. [24] Schnorrer F, Schönbauer C, Langer CCH, Dietzl G, Novatchkova M, Schernhuber K, et al. (2010). Systematic genetic analysis of muscle morphogenesis and function in *Drosophila*. Nature 464(7286):287–291. [25] Hassenstein B, Reichardt W (1956). Systemtheoretische Analyse der Zeit-, Reihenfolgen- und Vorzeichenauswertung bei der Bewegungsperzeption des Rüsselkäfers *Chlorophanus*. Z Naturforsch B 11(9–10):513–524. **[26] Ammer G, Serbe-Kamp E, Mauss AS, Richter FG, Fendl S, Borst A (2023). Multilevel visual motion opponency in *Drosophila*. Nat Neurosci 26(11):1894–1905.** [27] Tuthill JC, Wilson RI (2016). Mechanosensation and adaptive motor control in insects. Curr Biol 26(20):R1022–R1038. [28] Josephson RK (1985). Mechanical power output from striated muscle during cyclic contraction. J Exp Biol 114(1):493–512. [29] Walker SM, Schwyn DA, Mokso R, Wicklein M, Müller T, Doube M, Stampanoni M, Krapp HG, Taylor GK (2014). In vivo time-resolved

microtomography reveals the mechanics of the blowfly flight motor. PLoS Biol 12(3):e1001823. [30] Golan T, Raju PC, Kriegeskorte N (2020). Controversial stimuli: pitting neural networks against each other as models of human cognition. Proc Natl Acad Sci USA 117(47):29330–29337. [31] Hausen K (1982). Motion sensitive interneurons in the optomotor system of the fly. I. The horizontal cells: structure and signals. Biol Cybern 45(2):143–156. [32] Schnell B, Joesch M, Forstner F, Raghu SV, Otsuna H, Ito K, Borst A, Reiff DF (2010). Processing of horizontal optic flow in three visual interneurons of the *Drosophila* brain. J Neurophysiol 103(3):1646–1657. [33] Hill AV (1938). The heat of shortening and the dynamic constants of muscle. Proc R Soc Lond B 126:136–195. [34] Franklin DW, Wolpert DM (2011). Computational mechanisms of sensorimotor control. Neuron 72(3):425–442. [35] Dingemanse NJ, Dochtermann NA (2013). Quantifying individual variation in behaviour: mixed-effect modelling approaches. J Anim Ecol 82(1):39–54. **[36] Serbe-Kamp É, Bemme J, Pollak D, Mayer K (2023). Open Citizen Science: fostering open knowledge with participation. Res Ideas Outcomes 9:e96476. doi:10.3897/rio.9.e96476.** [37] Braitenberg V (1984). Vehicles: Experiments in Synthetic Psychology. MIT Press. **[38] Drews M, Leonhardt A, Pirogova N, Richter F, Schuetzenberger A, Serbe E, Braun L, Borst A (2020). Dynamic signal compression for robust motion vision in flies. Curr Biol 30(2):209–221.e8. doi:10.1016/j.cub.2019.10.035 [verify author order].** [39] Berg S, Beckett IR, Costa M, Schlegel P, Januszewski M, Marin EC, et al. (2026). Sexual dimorphism in the complete *Drosophila* male central nervous system connectome. Cell 189(18):5504–5526.e15. doi:10.1016/j.cell.2026.08.015 [verify and complete the author list as published]. [40] Matsliah A, Yu S-C, Kruk K, Bland D, Burke A, Gager J, et al. (2023). Codex: Connectome Data Explorer. Preprint, doi:10.13140/RG.2.2.35928.67844; codex.flywire.ai. **[41] Serbe-Kamp É (2026). MESH: connectome-scale contact-site detection and EM proofreading for putative gap junctions. Research software, MIT licence. github.com/ETigerschuss/MESH (development build); github.com/ActiveSensing/MESH (release v1.0.0 with pre-built viewer).** [42] Lawrence PA, Johnston P (1986). The muscle pattern of a segment of *Drosophila* may be determined by neurons and not by contributing myoblasts. Cell 45(4):505–513. [43] Usui-Aoki K, Ito H, Ui-Tei K, Takahashi K, Lukacsovich T, Awano W, Nakata H, Piao ZF, Nilsson EE, Tomida J, Yamamoto D (2000). Formation of the male-specific muscle in female *Drosophila* by ectopic *fruitless* expression. Nat Cell Biol 2(8):500–506. [44] Darwin Tree of Life Project. *Eristalis tenax* chromosome-level genome assembly idEriTena2.1 (GCA_905231855.1), Wellcome Sanger Institute, 2021; see also Hawkes W, Wotton K (2021). The genome sequence of the drone fly / tapered dronefly. Wellcome Open Res 6 [verify species naming and citation]. [45] Lappalainen JK, Tschopp FD, Prakhya S, McGill M, Nern A, Shinomiya K, Takemura S-y, Gruntman E, Macke JH, Turaga SC (2024). Connectome-constrained networks predict neural activity across the fly visual system. Nature 634:1132–1140. [46] Shiu PK, Sterne GR, Spiller N, Blanco-Redondo B, Alexander A, Takemura S-y, et al., Scott K (2024). A *Drosophila* computational brain model reveals sensorimotor processing. Nature 634:210–219 [verify author list]. [47] Schlegel P, Yin Y, Bates AS, Dorkenwald S, Eichler K, Brooks P, et al. (2024). Whole-brain annotation and multi-connectome cell typing of *Drosophila*. Nature 634:139–152. [48] Matsliah A, Yu S-C, Kruk K, Bland D, Burke AT, Gager J, et al. (2024). Neuronal parts list and wiring diagram for a visual system. Nature 634:166–180. [49] Bates AS, et al. (2026). Distributed control circuits across a brain-and-cord connectome. Nature, doi:10.1038/s41586-026-10735-w [verify volume and pages]. [50] Ammer G, Vieira RM, Fendl S, Borst A (2022). Anatomical distribution and functional roles of electrical synapses in *Drosophila*. Curr Biol 32(9):2022–2036.e4. [51] Juusola M, Dau A, Song Z, Solanki N, Rien D, Jaciuch D, Dongre SA, Blanchard F, de Polavieja GG, Hardie RC, Takalo J (2017). Microsaccadic sampling of moving image information provides *Drosophila* hyperacute vision. eLife 6:e26117. [52] Kemppainen J, Scales B, Razban Haghighi K, Takalo J, Mansour N, McManus J, Leko G, Saari P, Cooper J, Dau A, Yu H, et al., Juusola M (2022). Binocular mirror-symmetric microsaccadic sampling enables *Drosophila* hyperacute 3D vision. Proc Natl Acad Sci USA 119(15):e2109717119 [verify author list]. [53] Gallego G, Delbrück T, Orchard G, Bartolozzi C, Taba B, Censi A, Leutenegger S, Davison AJ, Conradt J, Daniilidis K, Scaramuzza D (2022). Event-based vision: a survey. IEEE Trans Pattern Anal Mach Intell 44(1):154–180. [54] He B, Wang Z, Zhou Y, Chen J, Singh CD, Li H, Gao Y, Shen S, Wang K, Cao Y, Xu C, Aloimonos Y, Gao F, Fermüller C (2024). Microsaccade-inspired event camera for robotics. Sci Robot 9(90):eadj8124 [verify author list]. [55] Paredes-Vallés F, Hagenaars JJ, Dupeyroux J, Stroobants S, Xu Y, de Croon GCHE (2024). Fully neuromorphic vision and control for autonomous drone flight. Sci Robot 9(90):eadi0591. [56] Hagenaars JJ, Paredes-Vallés F, de Croon GCHE (2021). Self-supervised learning of event-based optical flow with spiking neural networks. Advances in Neural Information Processing Systems 34 (NeurIPS 2021). [57] Gehrig M, Millhäusler M, Gehrig D, Scaramuzza D (2021). E-RAFT: dense optical flow from event cameras. International Conference on 3D Vision (3DV 2021), pp. 197–206. [58] Gehrig M, Aarents W, Gehrig D, Scaramuzza D (2021). DSEC: a stereo event camera dataset for driving scenarios. IEEE Robot Autom Lett 6(3):4947–4954. [59] Zhu AZ,

Thakur D, Özaslan T, Pfrommer B, Kumar V, Daniilidis K (2018). The Multivehicle Stereo Event Camera dataset: an event camera dataset for 3D perception. IEEE Robot Autom Lett 3(3):2032–2039. [60] Mitrokhin A, Ye C, Fermüller C, Aloimonos Y, Delbruck T (2019). EV-IMO: motion segmentation dataset and learning pipeline for event cameras. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2019), pp. 6105–6112. [61] Wiederman SD, Shoemaker PA, O'Carroll DC (2008). A model for the detection of moving targets in visual clutter inspired by insect physiology. PLoS ONE 3(7):e2784. [62] Nordström K, O'Carroll DC (2009). Feature detection and the hypercomplex property in insects. Trends Neurosci 32(7):383–391. [63] Nicholas S, Nordström K (2021). Facilitation of neural responses to targets moving against optic flow. Proc Natl Acad Sci USA 118(42):e2024966118 [verify]. [64] Neftci EO, Mostafa H, Zenke F (2019). Surrogate gradient learning in spiking neural networks: bringing the power of gradient-based optimization to spiking neural networks. IEEE Signal Process Mag 36(6):51–63. [65] Zenke F, Ganguli S (2018). SuperSpike: supervised learning in multilayer spiking neural networks. Neural Comput 30(6):1514–1541. [66] Deistler M, Kadhim KL, Pals M, Beck J, Huang Z, Gloeckler M, Lappalainen JK, Schröder C, Berens P, Gonçalves PJ, Macke JH (2025). Jaxley: differentiable simulation enables large-scale training of detailed biophysical models of neural dynamics. Nat Methods 22(12):2657–. doi:10.1038/s41592-025-02895-w [verify author list and pages]. [67] Davies M, Wild A, Orchard G, Sandamirskaya Y, Fonseca Guerra GA, Joshi P, Plank P, Risbud SR (2021). Advancing neuromorphic computing with Loihi: a survey of results and outlook. Proc IEEE 109(5):911–934. [68] Mayr C, Höppner S, Furber S (2019). SpiNNaker 2: a 10 million core processor system for brain simulation and machine learning. arXiv:1911.02385. [69] Pehle C, Billaudelle S, Cramer B, Kaiser J, Schreiber K, Stradmann Y, Weis J, Leibfried A, Müller E, Schemmel J (2022). The BrainScaleS-2 accelerated neuromorphic system with hybrid plasticity. Front Neurosci 16:795876. [70] Yao M, Richter O, Zhao G, Qiao N, Xing Y, Wang D, Hu T, Fang W, Demirci T, De Marchi M, Deng L, Yan T, Nielsen C, Sheik S, Wu C, Tian Y, Xu B, Li G (2024). Spike-based dynamic computing with asynchronous sensing-computing neuromorphic chip. Nat Commun 15:4464. [71] Collett TS, Land MF (1975). Visual control of flight behaviour in the hoverfly *Syritta pipiens* L. J Comp Physiol 99:1–66. [72] Collett TS, Land MF (1978). How hoverflies compute interception courses. J Comp Physiol 125:191–204. [73] Land MF, Collett TS (1974). Chasing behaviour of houseflies (*Fannia canicularis*). J Comp Physiol 89:331–357. [74] Hausen K, Strausfeld NJ (1980). Sexually dimorphic interneuron arrangements in the fly visual system. Proc R Soc Lond B 208:57–71. [75] Gilbert C, Strausfeld NJ (1991). The functional organization of male-specific visual neurons in flies. J Comp Physiol A 169:395–411. [76] Hornstein EP, O'Carroll DC, Anderson JC, Laughlin SB (2000). Sexual dimorphism matches photoreceptor performance to behavioural requirements. Proc R Soc Lond B 267:2111–2117. [77] Ribeiro IMA, Drews M, Bahl A, Machacek C, Borst A, Dickson BJ (2018). Visual projection neurons mediating directed courtship in *Drosophila*. Cell 174(3):607–621.e18. [78] Hindmarsh Sten T, Li R, Otopalik A, Ruta V (2021). Sexual arousal gates visual processing during *Drosophila* courtship. Nature 595:549–553. [79] Gailey DA, Taylor BJ, Hall JC (1991). Elements of the *fruitless* locus regulate development of the muscle of Lawrence, a male-specific structure in the abdomen of *Drosophila melanogaster* adults. Development 113(3):879–890. [80] Niven JE, Laughlin SB (2008). Energy limitation as a selective pressure on the evolution of sensory systems. J Exp Biol 211(11):1792–1804. [81] Yamins DLK, DiCarlo JJ (2016). Using goal-driven deep learning models to understand sensory cortex. Nat Neurosci 19(3):356–365. [82] Lindsey J, Ocko SA, Ganguli S, Deny S (2019). A unified theory of early visual representations from retina to cortex through anatomically constrained deep CNNs. International Conference on Learning Representations (ICLR 2019). [83] Cowley BR, Calhoun AJ, Rangarajan N, Ireland E, Turner MH, Pillow JW, Murthy M (2024). Mapping model units to visual neurons reveals population code for social behaviour. Nature 629:1100–1108. [84] Bajcsy R, Aloimonos Y, Tsotsos JK (2018). Revisiting active perception. Auton Robots 42:177–196. [85] Kim JS, Greene MJ, Zlateski A, Lee K, Richardson M, Turaga SC, Purcaro M, Balkam M, Robinson A, Behabadi BF, Campos M, Denk W, Seung HS, the EyeWirers (2014). Space–time wiring specificity supports direction selectivity in the retina. Nature 509:331–336. [86] Bonney R, Shirk JL, Phillips TB, Wiggins A, Ballard HL, Miller-Rushing AJ, Parrish JK (2014). Next steps for citizen science. Science 343(6178):1436–1437. [87] Marzullo TC, Gage GJ (2012). The SpikerBox: a low cost, open-source bioamplifier for increasing public participation in neuroscience inquiry. PLoS ONE 7(3):e30837. **[88] Madariaga D*, Arro D*, Irarrázaval C*, Soto A, Guerra F, Romero A, Ovalle F, Fedrigolli E, DesRosiers T, Serbe-Kamp É#, Marzullo TC# (2024). A library of electrophysiological responses in plants: a model of transversal education and open science. Plant Signal Behav 19(1):2310977. doi:10.1080/15592324.2024.2310977. *Equal contribution; #corresponding authors [verify the marks as published]. [89] Serbe-Kamp É, Aliyeva A, Krick CB, Jantus CI, et al. (2026). Gaze on IPCC climate maps carries behaviour-specific signals for intended low-carbon action. PsyArXiv, osf.io/preprints/psyarxiv/64utk (posted September 2026) [verify title, complete author list and order,**

**and DOI].** [90] Zhao A, Nern A, Koskela S, Dreher M, Erginkaya M, Laughland CW, Ludwigh H, Thomson A, Hoeller J, Parekh R, Romani S, Bock DD, Chiappe E, Reiser MB (2024). A comprehensive neuroanatomical survey of the *Drosophila* lobula plate tangential neurons with predictions for their optic flow sensitivity. eLife 13:RP93659 [verify author list]. [91] Borst A, Groschner LN (2023). How flies see motion. Annu Rev Neurosci 46:17–37. [92] Hürkey S, Niemeyer N, Schleimer J-H, Ryglewski S, Schreiber S, Duch C (2023). Gap junctions desynchronize a neural circuit to stabilize insect flight. Nature 618:118–125 [verify author list]. [93] Liu Z, Wang Y, Vaidya S, Ruehle F, Halverson J, Soljačić M, Hou TY, Tegmark M (2024). KAN: Kolmogorov–Arnold networks. arXiv:2404.19756; International Conference on Learning Representations (ICLR 2025) [verify venue]. [94] Gehrig M, Scaramuzza D (2023). Recurrent vision transformers for object detection with event cameras. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023), pp. 13884–13893 [verify pages]. **[95] Serbe E, Meier M, Leonhardt A, Borst A (2016). Comprehensive characterization of the major presynaptic elements to the *Drosophila* OFF motion detector. Neuron 89(4):829–841. [96] Leonhardt A, Ammer G, Meier M, Serbe E, Bahl A, Borst A (2016). Asymmetry of *Drosophila* ON and OFF motion detectors enhances real-world velocity estimation. Nat Neurosci 19(5):706–715. [97] Maisak MS, Haag J, Ammer G, Serbe E, Meier M, Kiefer M, Bahl A, Rubin GM, Nern A, Dickson BJ, Reiff DF, Hopp E, Borst A (2013). A directional tuning map of *Drosophila* elementary motion detectors. Nature 500(7461):212–216 [verify author list].** [98] Wehrhahn C (1979). Sex-specific differences in the chasing behaviour of houseflies (*Musca*). Biol Cybern 32:239–241. [99] Krapp HG (2015). How a fly escapes the reflex trap. Nat Neurosci 18(9):1192–1194. [100] Richards BA, Lillicrap TP, Beaudoin P, Bengio Y, Bogacz R, Christensen A, et al. (2019). A deep learning framework for neuroscience. Nat Neurosci 22(11):1761–1770. **[101] Bahl A, Serbe E, Meier M, Ammer G, Borst A (2015). Neural mechanisms for *Drosophila* contrast vision. Neuron 88(6):1240–1252 [verify pages]. [102] Haag J, Arenz A, Serbe E, Gabbiani F, Borst A (2016). Complementary mechanisms create direction selectivity in the fly. eLife 5:e17421. [103] Mauss AS, Meier M, Serbe E, Borst A (2014). Optogenetic and pharmacologic dissection of feedforward inhibition in *Drosophila* motion vision. J Neurosci 34(6):2254–2263.** [104] Wardill TJ, Fabian ST, Pettigrew AC, Stavenga DG, Nordström K, Gonzalez-Bellido PT (2017). A novel interception strategy in a miniature robber fly with extreme visual acuity. Curr Biol 27(6):854–859. [105] Franceschini N, Ruffier F, Serres J (2007). A bio-inspired flying robot sheds light on insect piloting abilities. Curr Biol 17(4):329–335. [106] Floreano D, Wood RJ (2015). Science, technology and the future of small autonomous drones. Nature 521(7553):460–466. [107] Bagheri ZM, Wiederman SD, Cazzolato BS, Grainger S, O'Carroll DC (2017). Performance of an insect-inspired target tracker in natural conditions. Bioinspir Biomim 12(2):025006 [verify]. [108] Gehrig D, Scaramuzza D (2024). Low-latency automotive vision with event cameras. Nature 629:1034–1040.